Harsh Patel

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SKILLS

Machine Learning Scikit-Learn, TensorFlow, AWS Sagemaker, Pandas, NumPy, Matplotlib, Seaborn

Languages Python, Javascript/Typescript, HTML, CSS, Java, C++, Processing

Tools Jupyter Notebook, Google Colab, REST APIs, Git, Github, VS-Code, UNIX Command-Line

Other Technical Documentation, Agile/Scrum Methodologies, Solution Architecture, Arduino

Soft Collaboration, Communication, Leadership, Time Management, Organization, Fast Learner

EXPERIENCE

Software Engineer - Friendlier Jan 2022 - Apr 2022

- Developed an automated payment and invoicing system using the Stripe API saving the company 32 hours per month
- Constructed firebase functions to track analytics within the application to seek areas of improvement and plan features
- Integrated Slack API to create a webhook directing help forms, removing the 1-hour delay between user and developer

Machine Learning Engineer Intern - Xerris Aug 2021

- Built technical documentation for clients and potential future developers outlining the 15 step instructions from preprocessing steps to model deployment through a combination of Jupyter Notebook and AWS Sagemaker
- Learned about Geometric Deep Learning and the math behind the concepts through manager-led training sessions
- Deconstructed machine learning python code into the various steps of the pipeline, we took into account how different variables throughout the code were storing training and test results to then be stored as a log of model progress

Software and Design Team Member - Robotics Team (FIRST Team 4733) Sep 2018 - Mar 2020

- Conceptualized the mechanical aspects of the block-carrying robot, problem-solving through 3 design requirements
- Monitored 300 lines of Java code for bugs on the software team, working directly with other members to resolve bugs
- Collaborated directly with team leads, actively learning by asking questions and observing the work done by all 5 teams

PROJECTS & COMPETITIONS

Dog Breed Image Convolution - Personal Project Apr 2022

- Implemented a TensorFlow image convolution model in order to make predictions about a dog's breed given the image
- Used libraries such as Pandas, Matplotlib, and Numpy to explore and build a stronger understanding of the dataset
- Utilized Google Colab to connect to a cloud-based GPU needed to train the model within a 30 minute time span

Heart Disease Classification / California Housing Regression - Personal Project Mar 2022

- Built two models, the first one being a classification model to determine whether someone has heart disease given 13 clinical parameters, the second a regression model useful for predicting California housing prices based on location
- Utilized libraries such as Scikit-Learn, Pandas, Matplotlib, Numpy, and Seaborn to analyze the datasets and train the models, also implemented practices such as hyperparameter tuning to further increase the accuracy of models
- Designed both models in Jupiter Notebook with comments and explanations for readers making it communicative

MIT COVID-19 Challenge - MIT Ideathon Nov 2020

- Oversaw the project by managing the creation of the idea and final presentation within a 36-hour window as team lead
- Presented all technical aspects of the project in a 5-minute presentation and answered any clarifying questions by judges
- Won 1st place for Systems involving Artificial Intelligence, out of a total of 20 other teams competing in the same track

EDUCATION

Candidate for B.A.S.c in Computer Engineering - University of Waterloo Sep 2021- Present